THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** small fish	Third level.
2	**(a)** oxygen	Photosynthesis.
3	**(d)** organic waste rots in it	Decomposition uses oxygen.
4	**(d)** foxes are removed	Fewer predators.
5	**(d)** global warming	Greenhouse effect.
6	**(d)** increase	Fewer predators.
7	**(c)** 50 J	5000 x 10% x 10%.
8	**(b)** burning of fossil fuels and deforestation	Human activity.
9	**(d)** stray dogs and diseases	Carcasses remain.
10	**(a)** more than one trophic level	Omnivore.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Energy decreases at each trophic level (1); Less energy is available to support them (1).	1+1
12	Kills useful insects as well (1); Passes along the food chain (1).	1+1
13	Not biodegradable (1); Animals swallow and get entangled (1).	1+1
14	Primary 2000 J, secondary 200 J (1); Tertiary = 20 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Algae: producer (1); Zooplankton: primary consumer (1); Fish and heron: secondary and tertiary consumers (1).	1+1+1
16	Deer population rises (1); Overgrazing destroys vegetation (1); Ecosystem degraded (1).	1+1+1
17	Collects rain from roofs (1); Recharges groundwater (1); Reduces water scarcity (1).	1+1+1
	OR: Frogs lose food and decrease (1); Snakes get less food from frogs (1); Plants may increase (1).	1+1+1
18	Plants trap silt (1); Microbes break down pollutants (1); Clean water goes to rivers and groundwater (1).	1+1+1
	OR: Take in CO2 (1); Transpiration adds vapour (1); Cool the area and bring rain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Grass to grasshopper to frog to snake to hawk (1); Grass to mouse to snake to hawk (1); Hawk eats mice and snakes (1); Without snakes mice and frogs increase (1); Hawks have fewer prey; grass is overgrazed (1).	1+1+1+1+1
	OR: Pond: algae, fish (1); Grassland: grass, herbivores (1); Decomposers present in both (1); Abiotic differs (1); Food chain examples (1).	1+1+1+1+1
20	Producer 20000 J (1); Herbivore 2000 J (1); Carnivore 200 J, top carnivore 20 J (1); 10 per cent rule assumption (1); Energy lost as heat limits chain length (1).	1+1+1+1+1
	OR: Pollinators: bees, butterflies (1); Seed dispersers: birds, squirrels (1); Symbionts: Rhizobium, lichens (1); Mutual benefits (1); Conservation of these species (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** stomata	Transpiration.
2	**(c)** less dissolved oxygen	Gas solubility decreases.
3	**(a)** producers	Pyramid of numbers.
4	**(c)** respiration	Oxidation of food.
5	**(c)** loss of plant cover and soil erosion	Land degrades.
6	**(b)** a frog eating insects	Eats a herbivore.
7	**(c)** nitrogen compounds	Rhizobium.
8	**(d)** using biodegradable material	Reduces pollution.
9	**(b)** They recycle nutrients to the soil	Recycling.
10	**(b)** restore soil nitrogen	Replenishes nitrogen.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Decompose dead matter into manure (1); Fix nitrogen in soil (1).	1+1
12	Reduce pollution and noise (1); Give shade and oxygen (1).	1+1
13	Plants release oxygen (1); Take in carbon dioxide (1).	1+1
14	Roots bind soil (1); Cover reduces the impact of rain (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Decompose only dead matter (1); Return nutrients to soil (1); Keep environment clean (1).	1+1+1
16	Species at risk of extinction (1); Tiger, snow leopard (1); Protected reserves (1).	1+1+1
17	Water pollution by run-off (1); Eutrophication (1); Harm to soil organisms (1).	1+1+1
	OR: Plants hold more energy than animals (1); Energy loses 90 per cent at every step (1); Eating plants is more efficient (1).	1+1+1
18	Oxides of sulphur and nitrogen dissolve in rain (1); Damage to crops and monuments (1); Water bodies become acidic (1).	1+1+1
	OR: Food and oxygen from plants (1); Pollination and seed dispersal by animals (1); Carbon dioxide from animals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Photosynthesis takes CO2 and gives oxygen (1); Respiration gives CO2 (1); Decomposition and burning give CO2 (1); Deforestation reduces CO2 uptake (1); CO2 rises, global warming (1).	1+1+1+1+1
	OR: Forests: oxygen, rain (1); Wetlands: water filtration (1); Oceans: food, climate control (1); Biodiversity (1); Sustainable use (1).	1+1+1+1+1
20	Nitrogen gas in air (1); Bacteria and lightning fix it (1); Plants absorb nitrates (1); Animals eat plants (1); Decomposers return nitrogen; legumes enrich soil (1).	1+1+1+1+1
	OR: Oxygen is used up by decay (1); Fishes die (1); Algal growth increases (1); Foul smell and disease (1); Remedy: waste treatment and awareness (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** filter water and support biodiversity	Natural filters.
2	**(c)** a food chain	Feeding relation.
3	**(b)** converting nitrogen to oxides	Natural fixation.
4	**(d)** storm surges and erosion	Natural barrier.
5	**(c)** plants and aeration	Photosynthesis.
6	**(b)** oxides of sulphur and nitrogen	Pollutants.
7	**(a)** rainwater harvesting	Saving water.
8	**(b)** dead matter will pile up and nutrients will not recycle	Cycle breaks.
9	**(b)** harmful ultraviolet rays	Absorbs UV.
10	**(d)** protecting trees	Hugging trees.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Competes with native species (1); May lack natural predators and spread (1).	1+1
12	Organisms have alternative food sources (1); Loss of one species does not collapse the system (1).	1+1
13	Diagram showing energy at each level (1); Energy decreases up the levels (1).	1+1
14	Causes air pollution (1); Destroys soil organisms and nutrients (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Animal respires giving out CO2 (1); Plant in light photosynthesises using CO2 (1); Balance can occur (1).	1+1+1
16	Little biodiversity (1); Pests spread fast (1); Soil nutrients are depleted (1).	1+1+1
17	Primary: deer (1); Secondary: wolf (1); Tertiary: eagle/lion (1).	1+1+1
	OR: They decompose kitchen and garden waste (1); Release nutrients (1); Improves soil structure (1).	1+1+1
18	Villagers hugged trees (1); Bishnois protect trees and animals (1); Community conservation (1).	1+1+1
	OR: Herbivores increase (1); Vegetation is overgrazed (1); Ecosystem collapses (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sewage, industrial waste (1); Fertiliser run-off (1); Dissolved oxygen falls (1); Fish die, disease spreads (1); Treatment plants, no dumping (1).	1+1+1+1+1
	OR: National parks: Corbett (1); Sanctuaries: Ranthambore (1); Reserves: Nilgiri (1); No hunting or cutting (1); Limited tourism (1).	1+1+1+1+1
20	Deforestation (1); Overfishing (1); Pollution (1); Urbanisation (1); Afforestation, conservation, awareness (1).	1+1+1+1+1
	OR: Save electricity and water (1); Use public transport (1); Reduce plastic (1); Segregate and compost waste (1); Plant trees (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.